Setting Up Jupyter Notebooks in VS Code

By the end of this guide, Visual Studio Code will be ready to open and run the course's Python notebooks.

This guide assumes that Python 3 and Visual Studio Code are already installed, and that the **Python** extension is installed in VS Code.

Step 1 — Install the Jupyter extension

1. Open VS Code.
2. Open **Extensions** from the left-hand activity bar, or use:
   - **Windows/Linux:** `Ctrl+Shift+X`
   - **macOS:** `Cmd+Shift+X`
3. Search for **Jupyter**.
4. Install the **Jupyter** extension published by **Microsoft**.
5. If VS Code asks you to reload the window, do so.

You should now have both the **Python** and **Jupyter** extensions installed.

Step 2 — Create and open a folder for the course

1. Create a folder somewhere easy to find, for example `database-notebooks`.
2. In VS Code, choose **File** → **Open Folder…** and select that folder.
3. Create a new notebook inside your folder:
   - In the **Explorer** tab, click the **New File…** icon next to your folder name
   - Type in a name followed by the `.ipynb` extension, for example `test.ipynb`

Keep the notebooks for this course in the same folder so that you can find them easily.

Step 3 — Create a virtual environment

A virtual environment is a separate place to install the Python packages. It avoids mixing them with packages used by other projects.

1. Open the Command Palette:
   - **Windows/Linux:** `Ctrl+Shift+P`
   - **macOS:** `Cmd+Shift+P`
2. Type `Python: Create Environment` and select that command.
3. Choose **Venv**.
4. Select the installed Python interpreter. If more than one is listed, choose a recent Python 3 version (3.10 or newer is recommended).

5. Let VS Code finish creating the environment. It will usually create a folder named `.venv` inside your course folder and select it for the workspace.

If VS Code asks which interpreter to use, choose the one inside your course folder's `.venv` environment—not a different project's environment.

Step 4 — Install the packages used by the notebooks

1. In VS Code, choose **Terminal** → **New Terminal**.
2. Check that the terminal is using the `.venv` environment. The command line should start with `(.venv)`. You can also check the Python interpreter shown in the VS Code status bar.
3. Run this command in the terminal, one at a time:

```
python -m pip install jupyter ipykernel pandas sqlalchemy "psycopg[binary]"
```

The same command works in the standard VS Code terminals on Windows, macOS, and Linux.

These packages provide:

- `jupyter` and `ipykernel`: notebook support and the Python kernel.
- `pandas`: tabular data analysis.
- `sqlalchemy`: database connections used by the Pandas notebook.
- `psycopg[binary]`: the Psycopg 3 PostgreSQL driver, including its required binary components for common supported systems.

You normally need to do this installation only once for this `.venv` environment.

Step 5 — Open a notebook and select the kernel

1. In the VS Code Explorer, open your notebook file (ending in `.ipynb`).
2. In the top-right corner of the notebook editor, click **Select Kernel**.
3. Choose **Python Environments** if VS Code asks for a category.
4. Select the Python interpreter inside the `.venv` environment you created for this course.

The selected kernel is the Python environment that actually runs the notebook's code. If you later open another course notebook, check that it uses the same `.venv` kernel.

Step 6 — Check that everything works

Create a temporary code cell in the notebook, paste the code below, and run the cell using the triangle **Run** button next to it.

```
import sys
import pandas as pd
import sqlalchemy
import psycopg

print("Python executable:", sys.executable)
print("Pandas:", pd.__version__)
print("SQLAlchemy:", sqlalchemy.__version__)
print("Psycopg:", psycopg.__version__)
```

If the cell runs without an error and the printed Python executable points to the course's `.venv` folder, the Jupyter environment is ready. You can delete the temporary cell afterwards.

If something does not work

- **`ModuleNotFoundError`:** Recheck that the notebook is using the course's `.venv` kernel. Then open a new VS Code terminal with that environment selected and run the package-installation command from Step 4 again.
- **The `.venv` environment is missing from the kernel list:** Open the Command Palette and run `Python: Select Interpreter`, choose the `.venv` interpreter, and reopen the notebook or use the notebook's **Select Kernel** menu again.
- **The `python` command is not found:** Check that Python is installed and detected by VS Code, then repeat Step 3 and select the correct Python interpreter.

Official references

- Jupyter Notebooks in VS Code
- Getting Started with Jupyter Notebooks in VS Code
- Python environments in VS Code